Exponential depth dependence in Bayesian neural-network regression

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Abstract

We construct bounded-weight tanh teachers of constant width and depth L for which a Gaussian Bayesian neural-network learner requires $\exp(\Omega(L))$ examples to generalize. Inputs follow one fixed nonatomic distribution on an arc of the unit circle, and training observes the teacher's real-valued outputs. The posterior conditions only on fitting these outputs within a fixed positive tolerance. In the strong-prior limit, and at every finite prior exponent $\gamma > 1$ at widths above a polynomial threshold in sample size and depth, the posterior fails to assign probability greater than $2/3$ to a correct prediction on at least half the test inputs, with sample probability at least $49/50$. The teacher iterates a fixed expanding map. Its outputs differ by a constant on exponentially close paired inputs. A sample of subexponential size, however, admits an exact interpolant with sufficiently low quadratic energy that the posterior concentrates on functions, or uniform approximations to functions, having much smaller Lipschitz constants. The pairing argument handles arbitrary posterior mixtures. All minimum-energy exact interpolants fail as well, including the specified iterated strong-prior, zero-tolerance posterior limits. Thus, for these learners, no sample bound that holds uniformly over nonatomic input distributions can be jointly polynomial in teacher width, weight bound, and depth. The fixed distribution is a Cantor measure; allowing the distribution to vary with depth also gives smooth arclength densities. Neither construction uses uniform measure on the circle. A separate width-first argument gives the same exponential obstruction at the mean-field endpoint $\gamma = 1$, for the same fixed input law, without a quantified width threshold. The original Boolean-cube classification problem remains outside these results.

1 Introduction

Can a Bayesian neural network learn every small bounded-weight teacher with a number of examples polynomial in both width and depth, without restricting the networks in its prior? On arbitrary continuous input distributions, the answer is negative for the Gaussian learners considered here. We give a family with fixed input dimension, fixed teacher width, fixed weight bound, fixed training and testing tolerances, and a single fixed input distribution. Only the teacher depth grows. Learning requires exponentially many examples in that depth.

The obstruction is a property of the learning rule, rather than an information-theoretic impossibility for every algorithm. A teacher can iterate a small expanding map, producing a highly oscillatory function. The independent Gaussian weight prior instead rewards a low-energy interpolant of the observed sample. At large learner width, such an interpolant can transmit a scalar signal through many layers with a small excess energy cost. Its product of layer norms, and hence its Lipschitz constant, is polynomial in the sample size. Below an exponential sample size this smoothness prevents it from approximating the teacher.

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The proof requires more than a large Lipschitz constant for one teacher implementation. We prove an approximation barrier for *every* function of small Lipschitz constant, exhibit a low-energy competitor for each typical sample, and transfer the resulting competition to the full posterior. A measure-preserving pairing of test inputs shows that mixing individually inaccurate predictions cannot repair the confidence guarantee.

We prove the strong-prior result first, then give a finite-scale extension for every fixed exponent above one. The latter uses Gaussian isoperimetry to remove a typical small-operator-norm Gaussian component. The actual posterior parameter vector need not itself have small Frobenius energy. We include the required memorization and perturbation arguments, so the proof does not require another paper from this research program. At the mean-field endpoint, a separate argument uses entropy costs and finite-dimensional large deviations. It establishes the same depth obstruction for the fixed nonatomic distribution as width tends to infinity, without a quantitative threshold on width.

2 Model and main conclusions

For input $x \in S^1 \subset \mathbb{R}^2$, write $\bar{x} = (1, x)$ and consider depth-L networks with $L-1$ hidden layers, all of width N:

$$h^0(x) = \bar{x}, \qquad h^j(x) = \tanh(W^j h^{j-1}(x)) \quad (1 \le j < L), \qquad z_w(x) = v^T h^{L-1}(x).$$

There are no separate internal biases. More generally, in input dimension d the independent centered Gaussian prior has variances

$$\mathrm{Var}(W^1_{ij}) = \sigma^2 N/d, \qquad \mathrm{Var}(W^j_{ab}) = \sigma^2 \ (2 \le j < L), \qquad \mathrm{Var}(v_a) = \sigma^2/N. \tag{1}$$

Its density in energy coordinates is proportional to $\exp[-\kappa_N(w)/(2\sigma^2)]$, where

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{j=2}^{L-1} \left\| W^j \right\|_F^2 + N \left\| v \right\|_2^2 . \tag{2}$$

The strong-prior limit means $\sigma \downarrow 0$ at fixed finite N, L. The finite exponent family sets $\sigma^2 = N^{-\gamma}$.

For a target f, independent sample inputs $X_i \sim D$, and observed real labels $Y_i = f(X_i)$, let $\Pi_{S,\sigma}$ be this prior conditioned on

$$|z_w(X_i) - Y_i| < \rho \qquad (1 \le i \le n). \tag{3}$$

There is no condition on unobserved inputs and no global output margin. Throughout the main results we fix

$$\rho = 1/8, \qquad \tau = 1/4,$$

and define the posterior's correct-output probability by

$$q_{S,\sigma}(x) = \Pi_{S,\sigma}\{|z_w(x) - f(x)| \le \tau\}.$$

Thus an input is a confidence error if $q_{S,\sigma}(x) \le 2/3$.

Here are explicit sample-dependent resource bounds used in the theorems:

$$\alpha = \frac{\log 2}{\log 20}, \quad \Delta_n = (100n^2)^{-1/\alpha}, \quad a_n = \Delta_n^{-1} \log(16n), \quad R_n = \max\{2, a_n\}, \tag{4}$$

$$M_n = \left\lceil \max\{4, n, 6nR_n^2\} \right\rceil , \qquad T_n = 4M_n n, \qquad K_n(L) = 2 + (L+14)T_n^{1/(L-1)}. \tag{5}$$

These bounds are fixed polynomials in n up to logarithmic factors. In particular M_n and T_n admit universal polynomial upper bounds.

Theorem 2.1 (Exponential lower bound in the strong-prior limit)**.** *There exist a fixed nonatomic Borel probability distribution D on S^1, targets $f_r : S^1 \to \mathbb{R}$ for all integers $r \geq 1$, and a universal integer r_0 with the following properties. Each f_r is implemented by a tanh teacher of width three, depth $L = r + 1$, and weights of absolute value at most 30. It satisfies $|f_r| \leq 13/20$ on the support of D. For every $r \geq r_0$, every integer $1 \leq n \leq e^{r/100}$, learner depth $L = r + 1$, and every fixed finite learner width $N \geq \max\{30, M_n\}$,*

$$\liminf_{\sigma \downarrow 0} \mathbb{P}_{S \sim D^n} \{D\{x : q_{S,\sigma}(x) \leq 2/3\} \geq 1/2\} \geq 49/50. \tag{6}$$

The distribution, input dimension, teacher width and weight bound, and both tolerances are independent of r, n, N.

Padding the teacher to width 30 makes its weights bounded by its width, if that is the chosen definition of teacher complexity. Thus its width-based complexity remains an absolute constant. The learner widths in Theorem 2.1 are allowed by any guarantee asserted for *every* width above teacher complexity. The theorem does not claim failure at every smaller width, including width equal to the teacher's.

Theorem 2.2 (Exponential lower bound at finite exponents above one)**.** *Use the distribution, targets and constants of Theorem 2.1. Fix $\zeta > 0$. There is a width threshold $\mathcal{N}_\zeta(n, L)$ with a polynomial upper bound in n, L, whose degree may depend on ζ, such that for every $r \geq r_0$, $1 \leq n \leq e^{r/100}$, $L = r + 1$, every $N \geq \mathcal{N}_\zeta(n, L)$, and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$,*

$$\mathbb{P}_{S \sim D^n} \{D\{x : q_{S,\sigma}(x) \leq 2/3\} \geq 1/2\} \geq 49/50.$$

In particular this applies at the actual finite scale $\sigma^2 = N^{-\gamma}$ for each fixed $\gamma > 1$, by taking $\zeta = \gamma - 1$. An explicit threshold is given in Lemma 6.2.

The same construction obstructs minimum-energy exact interpolation and a specified iterated Bayesian limit; see Section 8. At $\gamma = 1$, Theorem 9.1 gives a separate width-first lower bound: on a sample event of probability at least 49/50, the lower limit of the confidence-error mass as $N \to \infty$ is at least 1/2, for an exponential range of sample sizes. This endpoint theorem does not quantify how large the width must be. These conclusions are specific to the stated learner. They are not a lower bound for every method that knows the teacher family.

3 A constant-width expanding teacher

Define

$$J = [-3/4, 3/4], \qquad F(t) = 2\{\tanh(10(t + 1/2)) - \tanh(10(t - 1/2))\} - 1. \tag{7}$$

Lemma 3.1 (Two expanding branches)**.** *There are inverse branches $\phi_\pm : J \to J$ of F, with images in $[1/2, 13/20]$ and $[-13/20, -1/2]$, respectively, such that*

$$\frac{1}{20}|s - t| \leq |\phi_\pm(s) - \phi_\pm(t)| \leq \frac{1}{3}|s - t| \qquad (s, t \in J). \tag{8}$$

The two images have distance at least one.

Proof. The map F is even. On $[1/2, 13/20]$ it is strictly decreasing, and

$$F(1/2) = 2\tanh(10) - 1 > 3/4, \qquad F(13/20) < 1 - 2\tanh(3/2) < -3/4.$$

Moreover

$$20\{\operatorname{sech}^2(3/2) - \operatorname{sech}^2(10)\} \le |F'(t)| \le 20.$$

For explicit elementary bounds, the degree-four Taylor sum gives $e^3 > 15$, while the Taylor series and its geometric remainder give $e < 11/4$, hence $e^3 < 21$. These inequalities imply $\tanh(3/2) > 7/8$ and $\operatorname{sech}^2(3/2) > 1/6$, while $e^{20} > 20^4/24 > 4000$ gives $\operatorname{sech}^2(10) < 1/1000$. Thus the lower derivative bound exceeds three. The inverse function theorem gives the positive branch and its derivative bounds. Reflection gives the negative branch. Their image locations give the gap. □

Lemma 3.2 (Teacher implementation)**.** *For every $r \ge 1$, the function $f_r(x) = F^r(x_1)$ on S^1 has a width-three, depth-$(r+1)$ tanh implementation with weights bounded by* 30*.*

Proof. The first hidden layer computes

$$a_1 = \tanh(10x_1 + 5), \qquad a_2 = \tanh(10x_1 - 5), \qquad c = \tanh(\operatorname{artanh}(1/2)) = 1/2.$$

Every subsequent hidden layer applies

$$a_1' = \tanh(20a_1 - 20a_2 - 10c), \quad a_2' = \tanh(20a_1 - 20a_2 - 30c), \quad c' = \tanh(2\operatorname{artanh}(1/2)c) = 1/2.$$

If $2a_1 - 2a_2 - 2c = t$, the next two nonconstant neurons equal $\tanh(10(t+1/2))$ and $\tanh(10(t-1/2))$. The final readout $2a_1 - 2a_2 - 2c$ therefore gives $F^r(x_1)$. Every stated weight is at most 30 in absolute value, and the constant rail uses only the input constant and internal linear weights. □

4 A fixed input distribution and an approximation barrier

For independent fair signs $b_1, b_2, \ldots$, let

$$T = \lim_{j\to\infty} \phi_{b_1} \circ \cdots \circ \phi_{b_j}(0).$$

The contraction bound in Lemma 3.1 makes this limit well-defined. Let μ be its law and K its compact support. Then

$$K \subset [-13/20, -1/2] \cup [1/2, 13/20], \qquad F_*\mu = \mu.$$

Define $x(t) = (t, \sqrt{1-t^2})$ on $[-13/20, 13/20]$ and let $D = x_*\mu$. This is a Borel probability distribution on S^1. Since $\|x'(t)\|_2 = 1/\sqrt{1-t^2} < 2$ there, the embedding is two-Lipschitz. In particular the same D is used at every teacher depth.

Lemma 4.1 (Small balls and sample separation)**.** *The measure μ is nonatomic. For every interval I of length $0 < \ell \le 1$,*

$$\mu(I) \le 2\ell^\alpha, \qquad \alpha = \log 2/\log 20. \tag{9}$$

For n independent inputs from D, with probability at least $49/50$ their first coordinates are pairwise separated by at least Δ_n from (4).

Proof. Every sign word of length j specifies a cylinder of mass 2^{-j}. Distinct depth-j cylinders have distance at least $20^{-(j-1)}$: their first differing branch supplies a gap of one, and their common outer prefix contracts no distance by more than the lower bound in (8). The cylinder masses tending to zero show that μ has no atoms.

Put $j = \lfloor \log_{20}(1/\ell) \rfloor$. If $j \geq 1$, an interval of length ℓ meets at most one depth-j cylinder, because their separation is strictly larger than ℓ. Its mass is at most $2^{-j} \leq 2\ell^{\alpha}$. The case $j = 0$ follows from the trivial mass bound one. Endpoints do not alter this argument.

Conditioning on one of two independent first coordinates gives

$$\mathbb{P}\{|T_i - T_j| < \Delta_n\} \leq 2(2\Delta_n)^{\alpha}.$$

A union bound over unordered pairs is at most $2n^2\Delta_n^{\alpha} = 1/50$, since $2^{\alpha} \leq 2$. □

Lemma 4.2 (Paired-input barrier, including posterior mixtures)**.** *Let $r \geq 1$, $H \leq 3^r/10$, and $0 \leq h \leq 1/32$. Suppose a distribution on predictors assigns mass at least $1 - \beta$, where $\beta \leq 1/3$, to functions within uniform distance h on S^1 of some H-Lipschitz function. For target f_r and test tolerance $\tau = 1/4$, its correct-output probabilities satisfy*

$$D\{x : q(x) \leq 2/3\} \geq 1/2.$$

If the distribution is supported entirely on H-Lipschitz predictors, the conclusion also holds with $1/2$ in place of the confidence threshold $2/3$.

Proof. Let w be a uniform sign word of length r, let $Y \sim \mu$ be independent, and write ϕ_w for its composition. Pair

$$X_{\pm} = x(\phi_w(\phi_{\pm}(Y))).$$

A fair choice between these two points has law D, while

$$|f_r(X_+) - f_r(X_-)| = |\phi_+(Y) - \phi_-(Y)| \geq 1,$$

$$\|X_+ - X_-\|_2 \leq \frac{13}{5}\, 3^{-r}. \tag{10}$$

Thus an H-Lipschitz function changes by at most $13/50$ between the pair. A predictor within h of such a function cannot be correct at both points: that would force the approximating function's change to be at least $1 - 2(\tau + h) \geq 7/16 > 13/50$. For each pair, integration over the predictor distribution gives

$$q(X_+) + q(X_-) \leq 1 + \beta \leq 4/3.$$

At least one of the two probabilities is therefore at most $2/3$. Averaging the balanced pairing proves the claim. If every predictor is H-Lipschitz, the same sum is at most one and proves the final assertion. □

5 Exact memorization with a depth-independent Lipschitz ceiling

The next construction is useful because its energy grows almost linearly with depth, while its product norm has a bound independent of depth. It fits the observed real labels exactly.

Lemma 5.1 (Exact bounded-label memorization)**.** *Let $d, n \geq 1$, $L \geq 3$, and suppose sample inputs have projected coordinates $s_i = \langle u, x_i \rangle \in [-1, 1]$, for some unit vector u, with $|s_i - s_j| \geq \Delta > 0$ whenever $i \neq j$. For any labels $|y_i| \leq 7/8$, set*

$$a = \Delta^{-1} \log(16n), \quad R = \max\{2, a\}, \quad M = \left\lceil \max\{4, n, 3dnR^2\} \right\rceil, \quad T = 4Mn.$$

At every width $N \geq M$ there is a depth-L network with

$$z(x_i) = y_i, \qquad |z(x)| \leq 1 \quad (x \in \mathbb{R}^d), \qquad \kappa_N(w) \leq K := 2 + (L + 14)T^{1/(L-1)}.$$

Furthermore

$$\left(\frac{K+1}{L}\right)^L \le e^{17}T^{3/2}. \tag{11}$$

Proof. Order the coordinates as $s_1 < \cdots < s_n$ and put $c_j = (s_j + s_{j+1})/2$. For a vector $t \in \mathbb{R}^n$ define

$$g_t(x) = \frac{t_1 + t_n}{2} + \sum_{j=1}^{n-1} \frac{t_{j+1} - t_j}{2} \tanh(a(\langle u, x\rangle - c_j)). \tag{12}$$

The constant term is implemented using the hidden feature $1/2$ with first-layer bias $\operatorname{artanh}(1/2)$ and readout weight $t_1 + t_n$. Thus the hidden width is n, including when $n = 1$ and the sum is empty.

Write $\theta_j = \tanh(a(\langle u, x\rangle - c_j))$. Since the centers increase, $\theta_1 \ge \cdots \ge \theta_{n-1}$. Equation (12) is a convex combination of the t_i, with coefficients

$$\frac{1-\theta_1}{2}, \quad \frac{\theta_1 - \theta_2}{2}, \quad \ldots, \quad \frac{\theta_{n-2} - \theta_{n-1}}{2}, \quad \frac{1+\theta_{n-1}}{2}.$$

For $n = 1$ the sole coefficient is one. Hence $|g_t| \le \|t\|_\infty$ globally. Each sample is at distance at least $\Delta/2$ from every threshold, so $|\tanh v - \operatorname{sign} v| \le 2e^{-2|v|}$ gives

$$|g_t(x_i) - t_i| \le 2(n-1)e^{-a\Delta} \|t\|_\infty .$$

Let A be the linear map $t \mapsto (g_t(x_i))_{i=1}^n$. Then

$$\|A - I\|_{\infty\to\infty} < 1/8, \qquad \left\|A^{-1}\right\|_{\infty\to\infty} \le 8/7. \tag{13}$$

We next construct a scalar channel before selecting t. Put $m = L - 1 \ge 2$, $D_0 = 2T$ and

$$q = (D_0/4)^{1/(2m)} \ge 1, \qquad k_s = \min\{s, m-s\}, \qquad r_s = \frac{q^s}{\sqrt{D_0}\, k_s} \quad (1 \le s \le m-1).$$

These numbers satisfy $r_s \le 1/(2qk_s) \le 1/2$ and $r_{m-1} = 1/(2q)$. Define positive weights

$$\alpha_0 = \operatorname{artanh}(r_1), \qquad b_s = \frac{\operatorname{artanh}(r_s)}{r_{s-1}} \quad (2 \le s \le m-1),$$

and the scalar function

$$G(t) = 2q \tanh\big(b_{m-1} \tanh(\cdots \tanh(b_2 \tanh(\alpha_0 t)) \cdots)\big).$$

For $m = 2$ this means $G(t) = 2q \tanh(\alpha_0 t)$. By construction $G(1) = 1$. It is odd and strictly increasing, and is concave on $[0, 1]$ because positive rescaling and composition preserve increasing concave functions there. Thus G maps $[-1, 1]$ bijectively onto itself, $G(t) \ge t$ on $[0, 1]$, and $|G^{-1}(y)| \le |y|$ for $|y| \le 1$.

Choose

$$t = A^{-1}(G^{-1}(y_1), \ldots, G^{-1}(y_n))^T.$$

Equation (13) gives $\|t\|_\infty \le (8/7)(7/8) = 1$. Consequently $G \circ g_t$ has exact sample values y_i and is globally bounded by one.

We verify its energy. For $\|t\|_\infty \le 1$, the first matrix in (12) has squared Frobenius norm at most $3nR^2$. Its readout has squared norm at most $n + 3 \le 4n$. Pad with zero neurons to width M. The first-layer energy is then at most one, and the weighted readout energy at most T. For $N \ge M$, divide its N neurons into M balanced groups. Duplicate each old neuron within its group, and

divide its outgoing readout coefficient by the group size. Since each group size is between $N/(2M)$ and $2N/M$, this preserves the shallow function and gives first energy at most two and weighted readout energy $c^2 = N\,\|v_N\|_2^2 \le 2T = D_0$.

To implement G after that first layer, make every row of the next matrix equal to $\alpha_0 v_N^T$. Each subsequent scalar-channel matrix has all entries b_s/N, and the final readout has all entries $2q/N$. Their respective energy costs are $\alpha_0^2 c^2 \le D_0\alpha_0^2$, b_s^2, and $4q^2$.

The following elementary estimates bound their sum. The power series for artanh gives

$$(\operatorname{artanh}(v)/v)^2 \le 1 + v^2 \quad (0 \le v \le 1/2). \tag{14}$$

One verification is to bound $\operatorname{artanh}(v)/v \le 1 + v^2/[3(1-v^2)] \le 1 + 4v^2/9$; squaring this upper bound gives at most $1 + v^2$ for $v^2 \le 1/4$. If $R_s = k_{s-1}/k_s$, then

$$\sum_{s=2}^{m-1} R_s^2 \le m - 2 + 7, \qquad R_s \le 2, \qquad \sum_{s=1}^{m-1} k_s^{-2} < 4.$$

For the first inequality, pair the ratios $k/(k+1)$ and $(k+1)/k$ and use

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 \le 4/k^2,$$

whose sum is less than seven. A possible central ratio equals one. Equation (14) now gives

$$D_0\alpha_0^2 \le q^2 + 1/4, \qquad \sum_{s=2}^{m-1} b_s^2 \le q^2(m - 2 + 7) + 4.$$

The second remainder is bounded by $\sum_s q^2 R_s^2 r_s^2 \le \sum_s k_s^{-2} < 4$. Including a readout of size q would cost at most $(m+12)q^2$; the actual readout $2q$ adds $3q^2$. Including the first layer,

$$\kappa_N \le 2 + (m+15)q^2 = 2 + (L+14)(T/2)^{1/(L-1)} \le K.$$

Finally $K + 1 \le (L+17)T^{1/(L-1)}$ and $L/(L-1) \le 3/2$, which give (11). □

6 Posterior energy and output ceilings

Lemma 6.1 (Strong-prior ceiling)**.** *Fix a sample, a finite width N, depth $L \ge 3$, and positive tolerance ρ. Suppose it has an exact interpolant of energy at most K. Then the posterior defined by (1)–(3) satisfies*

$$\Pi_{S,\sigma}\{\kappa_N > K + 1\} \longrightarrow 0 \qquad (\sigma \downarrow 0).$$

If $K = 2 + (L+14)T^{1/(L-1)}$, it consequently assigns probability tending to one to functions of Lipschitz constant at most

$$\Lambda = \frac{e^{17/2}}{\sqrt{d}} T^{3/4}. \tag{15}$$

Proof. Continuity gives a parameter ball around the witness that still fits the sample within ρ and whose energies are at most $K + 1/4$. In the fixed finite parameter dimension its Gaussian probability has lower exponential rate at least $-(K+1/4)/2$ at speed σ^{-2}. The Gaussian radial tail outside energy $K+1$ has upper rate at most $-(K+1)/2$. These bounds follow directly by integrating the density over the ball and by the Gaussian squared-norm tail, respectively. Their quotient tends to zero.

For an energy at most $K+1$, arithmetic–geometric mean and the one-Lipschitz tanh activation give

$$\mathrm{Lip}(z_w)^2 \le \left\|W^1\right\|_F^2 \cdots \left\|W^{L-1}\right\|_F^2 \|v\|_2^2 \le \frac{1}{d}\left(\frac{K+1}{L}\right)^L \le \frac{e^{17}}{d}T^{3/2},$$

using (11). The constant input coordinate does not change the Lipschitz bound with respect to the actual input. □

At finite exponents above one we need a ceiling for a nearby function, rather than for the full parameter energy. We give its proof explicitly.

Lemma 6.2 (Finite-scale approximation ceiling)**.** *Suppose that for a sample on the Euclidean unit ball and every $N \ge M$ there is an exact interpolant of energy at most $K = 2 + (L+14)T^{1/(L-1)}$, where $L \ge 3$ and $T \ge 1$ are independent of N. Fix $\zeta > 0$, $0 < \beta < 1$, and $0 < h < \rho$. Define*

$$A = \left(\frac{K+2}{L-1}\right)^{(L-1)/2}, \qquad b^2 = \max\{2(d+1)\log 2 + 4\log 8,\ 16\log 9 + 8\log(8L)\},$$

$$\eta = \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{h}{\sqrt{2}LA}\right\}.$$

For every

$$N \ge \left\lceil \max\left\{M, 2, (b/\eta)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil, \qquad 0 < \sigma^2 \le N^{-(1+\zeta)}, \tag{16}$$

the posterior assigns mass at least $1-\beta$ to parameters w for which some network parameter U, depending on w, has $\kappa_N(U) \le K+1$, Lipschitz constant of z_U at most (15), and $\sup_{\|x\|_2 \le 1} |z_w(x) - z_U(x)| \le h$. The width threshold has a polynomial upper bound in $d, M, T, L, h^{-1}, \log(1/\beta)$ for fixed ζ.

Proof. The energy-coordinate matrices are

$$A_1 = \sqrt{d/N}W^1, \qquad A_j = W^j \ (2 \le j < L), \qquad A_L = \sqrt{N}v^T.$$

Their entries are independent $\mathcal{N}(0, \sigma^2)$ and their total squared Euclidean norm is κ_N. After normalizing the augmented input by $\sqrt{d}$ and hidden states by $\sqrt{N}$, the first activation map is $u \mapsto N^{-1/2}\tanh(\sqrt{N}A_1 u)$, and the other hidden activation maps have the same form. They are one-Lipschitz in their linear argument and zero-preserving. The normalized input norm on the unit ball is at most $\sqrt{2}$.

Suppose a residual network U has energy at most $K+1$, and a perturbation B has operator norm at most η in each normalized block. Telescoping one layer at a time and applying arithmetic–geometric mean to the other $L-1$ layer norms gives

$$\sup_{\|x\|_2 \le 1} |z_{U+B}(x) - z_U(x)| \le \sqrt{2}L\eta\left[\frac{K+1+2\eta\sqrt{L(K+1)} + L\eta^2}{L-1}\right]^{(L-1)/2}. \tag{17}$$

Indeed, writing the residual block norms as a_j, each telescoping term is at most $\sqrt{2}\eta$ times a product of $L-1$ numbers $a_j + \eta$. Their squared sum is bounded by the numerator, since $\sum a_j^2 \le K+1$ and $\sum a_j \le \sqrt{L(K+1)}$. The choice of η makes the cross term at most $1/2$ and $L\eta^2 \le 1/[16(K+1)]$. Thus that numerator is at most $K+2$ and (17) is at most $\sqrt{2}L\eta A \le h$.

Put $t^2 = N\sigma^2$. Let

$$a_d^2 = 2(d+1)\log 2 + 4\log 8, \qquad b_L^2 = 16\log 9 + 8\log(8L).$$

In energy coordinates let $\mathcal{T}$ be the set where first and last block Frobenius norms are at most ta_d, and internal block operator norms are at most tb_L. This set is nonempty, closed, convex and centrally symmetric, and its Gaussian probability is at least $1/2$. To verify the last assertion, the first squared norm has law $(t^2/N)\chi^2_{N(d+1)}$; the last uses N degrees of freedom. The exponential moment at parameter $1/4$ bounds either failure probability by $1/8$. A $1/4$-net of the unit sphere has at most 9^N points. Applying the scalar Gaussian tail on two such nets gives, for each internal matrix $tG/\sqrt{N}$,

$$\mathbb{P}\{\left\|tG/\sqrt{N}\right\|_{\mathrm{op}} > tb_L\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le 1/(4L).$$

The union bound proves the required mass. Every normalized block in $\mathcal{T}$ has operator norm at most tb.

Condition (16) gives $t \le N^{-\zeta/2}$, so $tb \le \eta$. If w_S is the exact-fitting witness, then $w_S + \mathcal{T}$ fits the sample within $h < \rho$ by the perturbation bound. The Gaussian shift identity and central symmetry give

$$\mu(w_S + \mathcal{T}) = e^{-\kappa_N(w_S)/(2\sigma^2)} \int_{\mathcal{T}} e^{-\langle w_S, B\rangle_\kappa/\sigma^2}\, d\mu(B) \ge \tfrac{1}{2} e^{-K/(2\sigma^2)}.$$

Pairing B with $-B$ in the integral proves the inequality.

Gaussian isoperimetry [1, 2], applied to a set of Gaussian mass at least $1/2$, gives

$$\mu\{\mathrm{dist}_\kappa(w, \mathcal{T})^2 > K + 1\} \le e^{-(K+1)/(2\sigma^2)}.$$

Dividing by the preceding lower bound for the conditioning event shows that the posterior probability of this event is at most $2e^{-1/(2\sigma^2)} \le \beta$. Off this event, let B be the metric projection of w onto $\mathcal{T}$ and set $U = w - B$. Then $\kappa_N(U) \le K + 1$, and (17) gives $\sup|z_w - z_U| \le h$. The product-norm bound gives the stated Lipschitz ceiling. No claim that w itself has low energy is needed.

Finally

$$K + 2 \le (L + 18)T^{1/(L-1)}, \qquad A^2 \le \left(\frac{L+18}{L-1}\right)^{L-1} T \le e^{19}T.$$

Thus every inverse radius and threshold in the lemma has the claimed polynomial upper bound for fixed ζ. □

7 Proof of the exponential lower bounds

For fixed $d = 2$, the quantities in (4)–(5) obey

$$T_n \le C_0 n^{4+4/\alpha}, \qquad \Lambda_n := \frac{e^{17/2}}{\sqrt{2}} T_n^{3/4} \le C_1 n^{3+3/\alpha}$$

for universal constants C_0, C_1. For instance use $\log(16n) \le Cn$ in the displayed definitions. The exponent $3 + 3/\alpha$ is less than 16. Hence there is a universal $r_0 \ge 2$ such that, for all $r \ge r_0$ and $n \le e^{r/100}$,

$$\Lambda_n \le 3^r/10. \tag{18}$$

No constant in this comparison depends on learner width or depth.

Proof of Theorem 2.1. The distribution and teachers were constructed in Sections 3–4. On the sample-separation event of Lemma 4.1, all labels belong to $K \subset [-13/20, 13/20]$, so Lemma 5.1 supplies an exact-fitting witness with energy at most $K_n(L)$ at every stated width. For each fixed

such sample and fixed width, Lemma 6.1 makes the posterior mass outside the Λ_n-Lipschitz class tend to zero. By (18), Lemma 4.2 applies as soon as that exceptional mass is at most $1/3$. Thus the indicator of the sample event inside (6) is eventually one for each separated sample. Fatou's lemma and the probability $49/50$ of separation prove the asserted lower limit. This use of Fatou does not require a small-noise threshold uniform over realized samples. □

Proof of Theorem 2.2. Use Lemma 6.2 with $K = K_n(L)$, $M = M_n$, $\beta = 1/4$ and $h = \rho/4 = 1/32$, increasing its width threshold to at least 30. On every separated sample the exact memorizer exists and the lemma gives posterior mass at least $3/4$ on outputs within h of Λ_n-Lipschitz functions. Equations (18) and Lemma 4.2 give the confidence-error conclusion. Separation has probability at least $49/50$. The explicit width threshold has a polynomial upper bound in n, L for fixed ζ, as proved in that lemma. □

8 Exact interpolation and the order of limits

Corollary 8.1 (Exact minimum-energy interpolants also fail)**.** *Under the parameters of Theorem 2.1, with sample probability at least* $49/50$*, every minimum-energy exact interpolant at every fixed width* $N \geq \max\{30, M_n\}$ *is* Λ_n*-Lipschitz and has regression error at least* $1/2$ *at tolerance* $1/4$*. Every mixture of such interpolants has confidence-error mass at least* $1/2$*, at confidence threshold* $2/3$ *or even* $1/2$*.*

At each such finite width, iterated weak limits of the posterior obtained by first sending $\sigma \downarrow 0$ *at fixed positive training tolerance and then sending that tolerance to zero exist subsequentially. Every such limit is supported on minimum-energy exact interpolants and obeys the same lower bounds.*

Proof. For a separated sample the exact memorizer has energy at most $K_n(L)$. The exact feasible set is nonempty and closed, and the quadratic energy is coercive. Its minimum therefore exists. Every minimizer has energy at most $K_n(L)$, so the arithmetic–geometric mean calculation in Lemma 6.1 bounds its Lipschitz constant by Λ_n. The pairing argument gives error at least $1/2$ for each function and the stated conclusions for all mixtures.

For completeness, fix the sample and width, and put

$$e_S(w) = \max_i |z_w(X_i) - Y_i|, \quad b_a = \inf_{\{e_S < a\}} \kappa_N, \quad b_0 = \min_{\{e_S = 0\}} \kappa_N.$$

An open ball within $\{e_S < a\}$ of energy arbitrarily close to b_a and the Gaussian tail estimate show that, as $\sigma \downarrow 0$, posterior energy concentrates at b_a. The measures are tight and every weak limit is supported on $\{e_S \leq a,\ \kappa_N = b_a\}$. In particular it is supported on the common compact set $\{\kappa_N \leq b_0\}$.

As $a \downarrow 0$, the numbers b_a increase to b_0. To verify the nontrivial inequality, take approximate minimizers at tolerances tending to zero. They lie in a compact energy sublevel, and a convergent subsequence has exact fit with energy at most $\lim_{a \downarrow 0} b_a$. Every outer weak limit is consequently supported on $\{e_S = 0, \kappa_N = b_0\}$. Compactness supplies the subsequential limits at both stages. □

The order of limits in the corollary is part of the statement. It does not identify a regular conditional posterior given exact real-valued observations at a fixed prior scale, or describe arbitrary simultaneous limits of the scale and tolerance.

9 An exponential lower bound at the mean-field exponent

At the critical exponent $\gamma = 1$, a different argument gives an exponential lower bound for the same teacher family and the same fixed nonatomic input distribution D. The width tends to infinity with teacher depth and the realized sample fixed. Only finite-dimensional large deviations are needed: apply them separately to the training sample and to one test pair, then integrate over the test pairing.

Theorem 9.1 (Critical regression lower bound)**.** *Use the distribution D and targets $f_h(x) = F^h(x_1)$ from Theorem 2.1. There are universal constants $c > 0$ and h_0 such that the following holds for every $h \geq h_0$ and integer $1 \leq n \leq e^{ch}$. Draw n independent inputs from D and observe their exact teacher outputs. Use the original Gaussian prior at depth $L = h + 1$, with first-layer entry variance $1/2$, internal entry variance $1/N$, and readout entry variance $1/N^2$. Condition solely on*

$$|z_w(X_i) - f_h(X_i)| < \rho, \qquad \rho = 1/8.$$

With sample probability at least $49/50$, the posterior satisfies

$$\liminf_{N\to\infty} \mathbb{E}_{w|S} D\{|z_w - f_h| > 1/4\} \geq \frac{1}{2}.$$

Moreover, writing $p_N(x) = \Pr_{w|S}\{|z_w(x) - f_h(x)| \leq 1/4\}$, the same samples obey

$$\liminf_{N\to\infty} D\{x : p_N(x) < 2/3\} \geq \frac{1}{2}.$$

The teacher still has width three and weights bounded by thirty. There is no global output-margin condition. These statements concern the actual posterior, including mixtures, and give no width threshold uniform in growing h.

9.1 Finite-domain rates and a metric bound

We include the large-deviation argument needed below. It works for any fixed finite input set $X = \{x^1, \dots, x^m\} \subset S^1$. For a feature law ν on $E = [-1, 1]^m$, write $Q(\nu) = \int hh^T \, d\nu(h)$. Let M_Q be the law of coordinatewise tanh applied to $\mathcal{N}(0, Q)$, and put

$$Q_0(x, y) = \frac{1 + x \cdot y}{2}, \qquad \mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^T Q^+ z, & z \in \operatorname{range} Q, \\ +\infty, & \text{otherwise.} \end{cases}$$

Here and below singular covariance matrices are allowed.

Lemma 9.2 (Finite-domain critical large deviations)**.** *The output vector $Z_N = (z_w(x))_{x\in X}$ obeys a large-deviation principle with speed N and good rate*

$$I_L(z) = \inf_{\nu_1,\dots,\nu_{L-1}} \left\{ \sum_{s=1}^{L-1} D(\nu_s \| \mathsf{M}_{Q_{s-1}}) + \mathcal{G}_{Q_{L-1}}(z) \right\}, \qquad Q_s = Q(\nu_s).$$

For every finite-cost path of total cost A in this formula,

$$|z(x) - z(y)| \leq \sqrt{A}\, |x - y| \exp\!\big(\sqrt{2(L-1)A}\big) \qquad (x, y \in X). \tag{19}$$

Proof. Let $\Gamma_m = \mathcal{N}(0, I_m)$, and take independent empirical measures $\eta_{s,N}$ of N samples from Γ_m, for $1 \le s < L$. Independently take $V_N = G/\sqrt{N}$, $G \sim \Gamma_m$. For $\mathsf{T}_Q(u) = \tanh(Q^{1/2}u)$, recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{Q(\nu_{s-1,N})})_{\#}\eta_{s,N}.$$

These are exactly the network's empirical hidden-feature laws on X: conditionally on the preceding layer, its next rows are independent Gaussian fields with that empirical second moment. The final output has exactly the law $Q(\nu_{L-1,N})^{1/2}V_N$.

Sanov's theorem and the Gaussian large-deviation principle give joint good rate $\sum_s D(\eta_s\|\Gamma_m) + \|v\|^2/2$. The recursive map just described is continuous in the weak topology for the measures and the ordinary topology for v. Indeed, matrix square roots are continuous; the bounded continuous test functions composed with T_Q converge uniformly on compact sets when Q varies; and weakly convergent measures form a tight sequence. Second moments on the compact feature space E are also continuous. The contraction principle therefore applies. The entropy-contraction identity

$$\inf_{\eta: T_{\#}\eta = \nu} D(\eta\|\Gamma_m) = D(\nu\|T_{\#}\Gamma_m)$$

follows by data processing and, for the reverse inequality, the lift $d\eta = q(T(u))\, d\Gamma_m(u)$ when $q = d\nu/d(T_{\#}\Gamma_m)$. Finally $\inf_{Q^{1/2}v=z} \|v\|^2/2 = \mathcal{G}_Q(z)$. These identities give the displayed rate, including the singular cases. This is an application of the standard Sanov and contraction theorems [3].

For the metric bound, write D_s for each hidden-layer entropy and r for the readout cost. Inverse tanh lifts a finite-entropy feature law to a law η with the same entropy relative to its reference Gaussian. The Gaussian quadratic transport inequality [4], applied after whitening on the covariance range, gives a coupling (U, G) satisfying

$$\mathbb{E}\|Q^{+1/2}(U - G)\|^2 \le 2D_s.$$

For $b = e_x - e_y$, Cauchy–Schwarz and Minkowski imply

$$\|b^T U\|_{L^2} \le (1 + \sqrt{2D_s})\sqrt{b^T Q b}.$$

Tanh is 1-Lipschitz, so this is also an upper bound on the next feature distance. The initial distance is $|x - y|/\sqrt{2}$. The readout inequality is $|b^T z| \le \sqrt{2r}\sqrt{b^T Q_{L-1} b}$. Multiply these inequalities and use

$$\sum_{s=1}^{L-1} \log(1 + \sqrt{2D_s}) \le \sqrt{2(L-1)\sum_s D_s} \le \sqrt{2(L-1)A}, \qquad r \le A.$$

This proves (19). □

Each test pair (X_+, X_-) from Lemma 4.2 has input distance at most $3 \cdot 3^{-h}$ and target gap at least one. Predicting both to tolerance $1/4$ requires an output gap of at least $1/2$. At $L = h + 1$, the metric bound shows that every such output on the two-point domain has

$$I_L(z) \ge h/100 \tag{20}$$

for all sufficiently large h: a path of cost $A \le h/100$ would instead give gap at most $3 \cdot 3^{-h}\sqrt{h/100}\exp(h/\sqrt{50})$, which tends to zero. The same conclusion holds after taking the infimum over paths.

9.2 A low-rate training interpolator

The next construction spends entropy only in the first layer and in logarithmically many final layers. Most hidden layers follow the uncontrolled Gaussian recursion.

Lemma 9.3 (A first-feature mean seed)**.** *Suppose $t_1 < \cdots < t_n$ belong to $[-2/3, 2/3]$ and have separation at least $\Delta \le 1$. For any signs y_i and any fixed finite domain containing $x(t_i)$, a genuine first-feature law ν satisfies*

$$D(\nu\|\mathsf{M}_{Q_0}) \le D, \qquad \mathbb{E}_\nu H(x(t_i)) = m_n y_i, \qquad m_n = \frac{1}{2n},$$

where $1 \le D \le C\Delta^{-2}[1+\log(n+1)]^2$ for a universal C.

Proof. Use the ideal step vectors $s_0 = 1$ and $s_j(i) = \operatorname{sign}(t_i - (t_j + t_{j+1})/2)$, $1 \le j < n$. The inverse of this basis maps the maximum norm to coefficient ℓ^1 norm with norm at most n: the coefficients are successive half-differences and the half-sum of the two endpoint values.

In first-layer weight space, shift the bias and first coordinate so that a row's preactivation mean is $B(t-u_j)$, where u_j is the corresponding midpoint. For the constant vector use constant mean B. The random preactivation variance is one on S^1. Since

$$1 - \mathbb{E}\tanh(m+Z) \le 2e^{-2m+2} \qquad (Z \sim \mathcal{N}(0,1)),$$

choosing $B = C\Delta^{-1}[1+\log(n+1)]$ makes the resulting mean vectors v_j approximate s_j to maximum error $1/(4n)$. A Neumann series gives exact coefficients $y = \sum_j c_j v_j$ with $\sum_j |c_j| \le 4n/3$. Mix the corresponding signed Gaussian shifts with weights $m_n|c_j|$, and put the remaining probability on the unshifted Gaussian law. Negating a shift negates its feature mean by oddness. The desired mean identity follows. Each shift has Gaussian relative entropy at most CB^2, and convexity and data processing give the stated bound. These are laws of actual weight vectors, so their feature means and covariances are defined on the entire finite domain, not only on the sample. □

Lemma 9.4 (Free cooling and late amplification)**.** *Suppose a first-feature law has entropy at most $D \ge 1$ and mean $a_0 y_i$ at selected inputs, where $y_i \in \{-1, 1\}$ and $0 < a_0 \le 1$. For every fixed $0 < \eta < 1$, there are constants C_1, C_2, depending only on η, such that if*

$$L \ge C_1[1 + \log(D/a_0^2) + \log(L+1)],$$

then the critical rate admits an output z with

$$|z(x_i) - a y_i| < a\eta, \qquad I_L(z) \le C_2[1 + \log(D/a_0^2) + \log(L+1)], \qquad a = 3/5.$$

Proof. Mix the supplied law with the reference first-feature law at weight $1/D$. This costs at most one and yields a covariance $Q \succeq bb^T$ with $b(x_i) = a_0 y_i/\sqrt{D}$. The vector b is the full-domain mean of the supplied law divided by $\sqrt{D}$.

Consider an uncontrolled transition with $Q \succeq bb^T$ and $q_x = Q(x,x) \le 1$. Write the reference Gaussian field as $bZ_0 + G_R$. Project its tanh features onto Z_0. Gaussian integration by parts shows that the resulting covariance dominates $b'b'^T$, where

$$b'_x = b_x h(q_x), \qquad h(q) = \mathbb{E}\operatorname{sech}^2(\sqrt{q}Z) = 1 - F_0(q), \qquad F_0(q) = \mathbb{E}\tanh^2(\sqrt{q}Z).$$

The new diagonal is $F_0(q_x)$. There are universal $c > 0, C < \infty$ such that

$$F_0(q) \le q - cq^2, \qquad -\log h(q) \le Cq \quad (0 \le q \le 1).$$

For example $c = 1/100$ works. On $|u| \le 1$, $u^2 - \tanh^2 u \ge u^4/6$; integrate over $|Z| \le 1$ to obtain the first bound. For the second, $h(q) \ge h(1) \ge 1/8$ and $F_0(q) \le q$. It follows that after H free layers the diagonal is at most $(1 + cH)^{-1}$ and the signed seed at every selected input is at least

$$u_* := e^{-8}(L+1)^{-800} a_0/\sqrt{D} \qquad (H \le L). \tag{21}$$

Indeed the diagonal reciprocals grow by at least c per step, and summing $-\log h(q) \le 8q$ proves this loose explicit bound.

Now suppose $Q \succeq bb^T$ and $y_i b_i \ge u > 0$ on the sample. Shift the reference Gaussian by Ab. The entropy cost is at most $A^2/2$, since $b^T Q^+ b \le 1$. The new mean b' satisfies

$$y_i b'_i \ge c_0 \min\{Au, 1\}, \qquad c_0 = 1/100,$$

and the new covariance dominates $b'b'^T$. To check the inequality, integrate the derivative $\mathbb{E}\operatorname{sech}^2(v + \sqrt{q}Z)$ for $0 \le v \le 1$ and $q \le 1$, restricting to $|Z| \le 1$; larger means follow by monotonicity. Choose $A = 2/c_0$. Repeating this shift doubles the minimum signed mean until it reaches $c_0/2$, and preserves this lower bound thereafter.

Choose

$$m = \max\{0, \lceil \log_2(c_0/(2u_*)) \rceil\}, \qquad H = L - m - 3.$$

The stated depth condition, with a sufficiently large C_1, ensures $H \ge 0$. Use one initial seed layer, H free layers, m shift layers, and one final saturation layer. This is exactly $L - 1$ hidden layers and avoids any circular choice of the cooling time. For the final shift use Bb, with B large enough in terms of η that its feature mean v obeys $|v_i - y_i| < \eta$. The bound

$$1 - y_i \mathbb{E} \tanh(G_i + Bb_i) \le 2e^{2 - Bc_0}$$

suffices. Its entropy cost is at most $B^2/2$. Finally $Q_{L-1} \succeq vv^T$, so $z = av$ has readout cost at most $a^2/2$. The total cost is at most $1 + mA^2/2 + B^2/2 + a^2/2$, which has the asserted bound. Every Gaussian shift and rank-one domination is on the full finite domain throughout the construction. □

9.3 Comparison of posterior masses

Proof of Theorem 9.1. By Lemma 4.1, the training scalar coordinates are Δ_n-separated with probability at least $49/50$, where $\Delta_n = (100n^2)^{-1/\alpha}$ and $\alpha = \log 2/\log 20$. Fix any such sample. Since the Cantor measure is invariant under F, every teacher label belongs to $[-13/20, -1/2] \cup [1/2, 13/20]$. Set $y_i = \operatorname{sign} f_h(X_i)$ and $a = 3/5$. Thus

$$|f_h(X_i) - ay_i| \le 1/10.$$

Apply Lemma 9.3 on the finite training set. Its quantity D/m_n^2 is polynomial in n. Lemma 9.4, with $\eta = 1/100$, then supplies an output whose training errors are less than

$$1/10 + a\eta = 53/500 < 1/8.$$

Reduce the universal c and increase h_0 so that, whenever $n \le e^{ch}$, the depth condition holds and this witness's rate is less than $h/200$. The open training event A_S therefore satisfies

$$\liminf_{N \to \infty} \frac{1}{N} \log \Pr(A_S) > -h/200. \tag{22}$$

The strict inequality follows from the strict witness cost bound.

Now fix one test pair (X_+, X_-) in the balanced pairing of Lemma 4.2. Let B_{X_+,X_-} be the closed event that both real outputs are within $1/4$ of their teacher values. Apply the two-point version of Lemma 9.2. Equation (20) gives

$$\limsup_{N\to\infty} \frac{1}{N} \log \Pr(B_{X_+,X_-}) \le -h/100.$$

Together with (22), the elementary ratio bound

$$\Pr_{w|S}(B_{X_+,X_-}) \le \frac{\Pr(B_{X_+,X_-})}{\Pr(A_S)}$$

shows that this posterior probability tends to zero for every fixed test pair. No joint or functional large-deviation principle is being assumed. The width at which the ratio becomes small may depend on the test pair.

Let $b_N(X_+, X_-)$ denote this pair-correct probability and average with respect to the pairing law. Bounded convergence gives $\mathbb{E}b_N \to 0$. Inclusion–exclusion gives, pair by pair,

$$p_N(X_+) + p_N(X_-) \le 1 + b_N(X_+, X_-).$$

A fair choice between the paired inputs has law D. Averaging proves that the posterior Gibbs error is at least $(1 - \mathbb{E}b_N)/2$, giving its asserted lower limit.

For the confidence assertion, if $b_N < 1/3$, both correct-output probabilities cannot be at least $2/3$. At least one member of the pair has probability strictly below $2/3$. Consequently

$$D\{x : p_N(x) < 2/3\} \ge \frac{1}{2} \Pr_{\text{pair}}\{b_N < 1/3\} \ge \frac{1 - 3\mathbb{E}b_N}{2}.$$

Taking lower limits proves the confidence claim, since $\mathbb{E}b_N \to 0$. The argument applies for every separated sample and handles arbitrary posterior mixtures. □

10 Smooth input densities when the distribution may vary with depth

Corollary 10.1 (Smooth-distribution variant)**.** *For every r there is a probability distribution D_r on S^1 with a smooth density with respect to arclength such that Theorems 2.1 and 2.2 and Corollary 8.1 hold with D_r replacing D. The teacher f_r, sample range, resource bounds and confidence-error conclusions are unchanged. The width-first mean-field conclusion of Theorem 9.1 also holds, with universal constants of the same kind, which may be chosen to serve both statements. No uniform bound on the density or its derivatives is asserted as r grows.*

Proof. Let U_r have a smooth compactly supported probability density on $[-10^{-2}20^{-r}, 10^{-2}20^{-r}]$, independently of $T \sim \mu$, and let D_r be the law of $x(T + U_r)$. The scalar law is the convolution of μ with that smooth density. Its support is in $(-.66, .66)$, so its pushforward has a smooth arclength density on the circle, extended by zero off this arc.

For every interval I, conditioning on U_r preserves the bound $\mathbb{P}(T + U_r \in I) \le 2|I|^\alpha$. Thus the sample-separation probability and all resources remain unchanged. Globally $|F'| \le 20$, so

$$|F^r(t + u) - F^r(t)| \le 20^r|u| \le 1/100.$$

The real training labels therefore have magnitude at most $13/20 + 1/100 = .66 < 7/8$, as required by Lemma 5.1.

Use the same jitter U_r for both members of each pair in Lemma 4.2. The scalar input gap is unchanged, the arc embedding remains two-Lipschitz, and the target gap is at least $1 - 2/100 = .98$. For the test tolerance and every $h \le 1/32$, the lower bound on an approximating function's change is now $.98 - 2(1/4 + h) \ge .4175$, still greater than $13/50$. The same pairing and posterior arguments therefore apply.

For the mean-field statement, the smoothed real label differs from $3/5$ times its sign by at most .11. The low-rate witness in Theorem 9.1 adds less than .006 training error, so its total error is less than $.116 < 1/8$. The first-feature seed applies on the slightly enlarged arc and uses the unchanged sample separation. Correct prediction of both members of a smoothed pair requires an output gap of at least $.98 - 2(1/4) = .48$. The same entropy-based metric bound excludes this at rate at most $r/100$ for large r. The finite-pair posterior ratio and bounded-convergence argument then prove the mean-field conclusion as well. □

11 Scope and interpretation

The construction rules out a general guarantee jointly polynomial in teacher width, bounded weights and depth for the stated Gaussian learners on arbitrary continuous-domain distributions. Given any proposed joint polynomial, one can choose depth large enough that its sample requirement is below $e^{(L-1)/100}$ and then choose an allowed learner width from the theorem. In the strong-prior case the failure persists arbitrarily far into the small-scale limit. At every fixed exponent above one it already occurs at the actual finite prior scale, for sufficiently large polynomial learner width.

The distributional qualification is substantial. The fixed input law is nonatomic but singular: it is supported on a Cantor subset of a circle arc. Smooth arclength densities are possible when the input law may vary with depth, by Corollary 10.1. This paper does not prove the same lower bound for uniform measure on the circle or sphere, for one fixed smooth input density, or for every smooth density. Nor does it claim a lower bound for every possible learning algorithm. Its proof uses the specific energy competition induced by the Gaussian prior.

The continuous domain matters. It contains distinct points at exponentially small Euclidean distances. Distinct Boolean-cube vertices have distance at least two, so this close-pair mechanism does not transfer to the original Boolean classification model. A coordinate restriction or a grid encoding by Boolean strings also changes the network representation and its norm. Consequently this result leaves open the general small-deep-circuit question on the Boolean cube.

The Gaussian-distance proof requires $\gamma > 1$ because the typical Gaussian component then has operator norm tending to zero with width. The separate mean-field proof instead uses finite-dimensional entropy costs and takes width to infinity without a quantified threshold. At this endpoint, the positive sample exponent need not equal $1/100$, and the one-half confidence-error bound is a lower limit in width rather than a finite-width assertion. The theorem concerns increasing *teacher* depth, and does not contradict polynomial dependence on additional learner depth for a fixed shallow target. No claim about non-Gaussian priors is made here.

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